CHAPTER 1
INTRODUCTION

Background of the Study

In Colegio de San Jose, lost and found items are currently handled manually through the school office. Students who lose their belongings need to personally visit the office to check if their items have been found. While this system works, it can be time-consuming, unorganized, and sometimes inconvenient, especially when students are unsure whether their item has already been reported. In addition, manual verification of item ownership may create risks of misclaiming or lack of proper documentation.

According to Algurni (2021) web-based platforms improve organizational efficiency by enabling real-time access to centralized data and supporting decision-making through improved information flow and coordination among stakeholders. These systems also enhance transparency and accountability by providing structured digital records that are easier to manage compared to manual documentation. In the context of user acceptance, contemporary studies on technology adoption indicate that users are more likely to embrace information systems when perceived usefulness and ease of use positively influence their intention to use the system (Xue, Rashid, & Ouyang, 2024).

As ICT students, the researchers believe that technology can improve the efficiency, organization, and security of this process. For this reason, the

researchers developed iFind, a web-based Lost and Found System with user authentication and claim verification features. The system allows registered users to log in using accounts created by the administrator. Students and staff can submit reports for lost or found items, which are categorized and stored in a database. The platform includes a search functionality that enables users to quickly find specific items. To improve security and prevent false claims, the system requires users to provide proof of ownership before claiming an item. The administrator verifies the submitted proof before marking the item as officially claimed. This ensures accountability and protects both the claimant and the school.

Consequently, the development of the iFind: Lost and Found System incorporates key features such as authentication, categorized listings, search functionality, and claim verification, all of which are designed to be intuitive and supportive of users' needs. These features enable users to easily report, search, and claim lost items within a structured and secure environment. Furthermore, usability research indicates that clear navigation, responsive design, and intuitive interfaces significantly enhance user experience and task efficiency in web-based applications. Guided by these principles, the iFind system was developed with a strong emphasis on accessibility, ease of use, and user satisfaction. Ultimately, the system seeks to improve the school's existing lost and found processes by providing a more accessible, organized, secure, and transparent platform through a structured web-based solution.

Theoretical Framework

The study is supported by contemporary research on user acceptance of information systems using the Unified Theory of Acceptance and Use of Technology (UTAUT), which integrates multiple determinants of user intention and technology usage behavior in educational contexts (Xue, Rashid, & Ouyang, 2024). The UTAUT framework highlights that performance expectancy, effort expectancy, social influence, and facilitating conditions significantly influence users' intention to adopt and use new technology platforms. By aligning with these determinants, the design of iFind prioritizes ease of use, usefulness, and accessibility to encourage adoption among students and staff.

Conceptual Framework

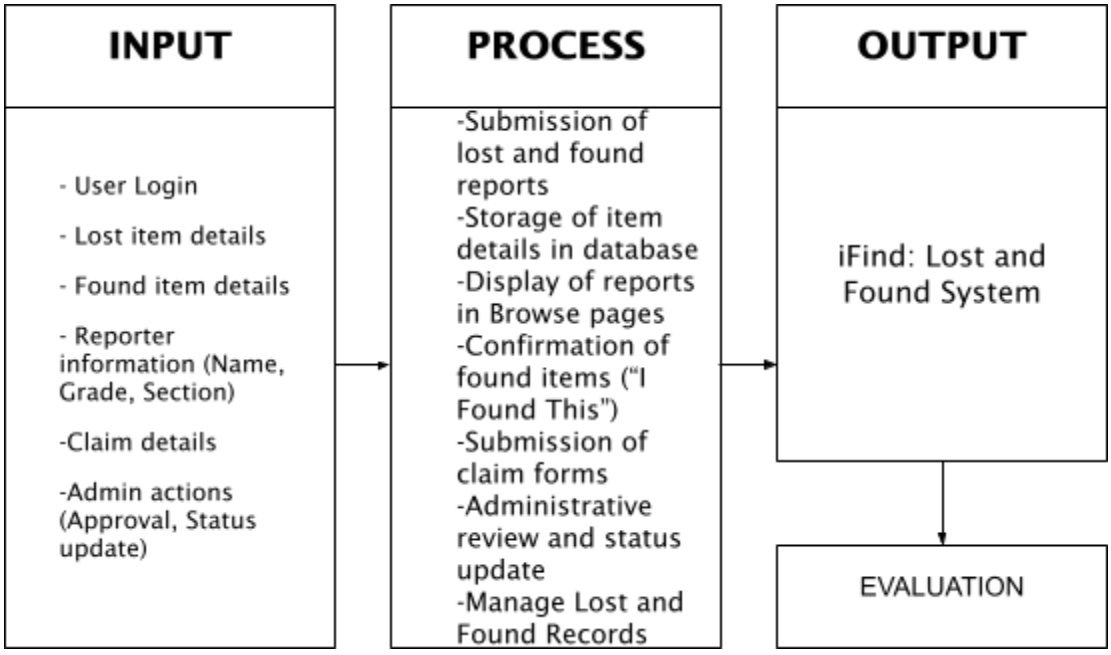


Figure 1. Conceptual Framework of the Study

The conceptual framework of the study is shown in Figure. 1. the Input consists of user account information, lost and found item details (including descriptions, images, and categories), and claim verification proof submitted by users. The Process involves user authentication through administrator-created accounts, storing item records in the database, organizing items by category, enabling search functionality, verifying claim proofs submitted by users, and updating item status (lost, found, claimed). The Output is the iFind system, a secure web-based Lost and Found platform that allows authenticated users to report, browse, search, and claim items while enabling administrators to verify claims and manage records efficiently.

Objectives of the Study

The general objective of this study is to develop **iFind**, a web-based lost and found system that helps students and school staff report, search, and manage lost items in a more organized and convenient way in Colegio de San Jose.

Specifically, the study aims to:

- manage user accounts
- manage lost and found items
- create user and admin notification messages
- generate reports

- evaluate the system based on ISO 25010's eight (8) software quality standards namely Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Definition of Terms

Administrator. A system administrator (sysadmin) is an information technology professional who supports a multiuser computing environment and ensures continuous, optimal performance of IT services and support systems. (Katie Terrell Hanna, 2021)

In this study, the Administrator refers to the authorized personnel who manages the iFind system by monitoring lost and found reports, verifying item ownership claims, and updating the status of items within the system.

Agile Development Model. Agile software development is an approach to software development that emphasizes iterative development, collaboration, and the ability to respond to change quickly (Beck et al., 2001).

In this study, the Agile Development Model refers to the approach used in developing the iFind system where features are created, tested, and improved through several development cycles.

ISO/IEC 25010 is an international standard that defines a software product quality model and identifies characteristics used to evaluate software quality (ISO/IEC, 2011).

In this study, ISO/IEC 25010 refers to the international standard used to evaluate the quality of the iFind system based on selected software quality characteristics.

Lost and Found System. A lost and found system is a service designed to assist individuals in recovering items they have lost, typically involving a designated location where lost belongings are securely stored until claimed by their rightful owners (Ahmad et al, 2019).

In this study, the Lost and Found System refers to the iFind platform developed for Colegio de San Jose to help manage and track lost and found items within the school.

User Authentication. User authentication refers to the process of verifying the identity of a user attempting to gain access to a computer network, system or device. (Rahul Awati, 2024)

In this study, User Authentication refers to the login process that allows only registered users to access the iFind system.

Web-Based System. A web-based system is an application that is accessed through a web browser over a network such as the internet (Kroenke & Boyle, 2019).

In this study, a Web-Based System refers to an application that can be accessed through a web browser using an internet connection.

Significance of the Study

This study is significant for the following peoples:

Students. The system will help students easily report and track their lost belongings, reducing the time and effort needed to recover them. It also promotes responsibility and trust in using technology for school-related concerns.

School Staff. School staff can use the system to record and monitor found items more efficiently, making the process more organized compared to traditional manual recording. This reduces confusion and ensures accurate tracking of lost and found items.

School Administration. The administration benefits from automated reports and reliable records that can be used for monitoring and decision-making. This strengthens transparency and accountability in handling student concerns.

ICT Community. The project contributes to the ICT community by showcasing the application of web technologies in solving real-world school problems. It also highlights the potential of system development in improving organizational processes.

Future Researcher. The system will serve as a useful reference for future researchers who plan to develop or improve digital systems for tracking and recovering lost items. It provides insights into the design, implementation, and effectiveness of a centralized platform that facilitates efficient reporting and retrieval of lost belongings. Future researchers may also use this study as a basis for enhancing the system by integrating advanced technologies and applying it in other institutions or

Scope and Limitation of the Study

Scope

The study focuses on the design and development of the iFind System, a secure web-based Lost and Found Information System intended for use at Colegio de San Jose. The system includes user authentication, where accounts are created and managed by the administrator. Only registered users can access the system features. The system allows students and staff to submit reports for lost or found items, upload images, assign item categories, and browse or search reported items through a search bar. Submitted items are stored in a database and displayed on the website with their corresponding details and status (lost, found, or claimed). A claim verification feature is included, requiring users to provide proof of ownership before claiming an item. The administrator reviews and verifies the submitted proof before updating the item status to

"claimed." This ensures security and prevents unauthorized claims. The platform is accessible through web browsers using devices such as computers or mobile phones with internet access.

Limitation

The study is limited to the development of a web-based system and does not include a mobile application or offline functionality. The system requires an internet connection to access and use its features. The system only handles lost and found items within the premises of Colegio de San Jose. It does not accommodate users outside the school community. The claiming of items is not processed online; it must still be verified and completed at the school office.

Additionally, the system does not include advanced features such as automated item matching, push notifications, image recognition, or integration with other school systems. The study focuses only on providing a functional and organized digital platform for reporting and browsing lost and found items. Testing and evaluation of the system are limited to selected students, staff, and administrators. Large-scale implementation and future improvements are beyond the scope of this study.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Related Literature

Record Keeping Systems

Traditional record-keeping systems in many institutions rely on manual documentation such as logbooks and paper-based files. Manual processes are often inefficient, prone to human error, and difficult to manage as records increase. Recent studies emphasize that organizations transitioning from manual to digital information systems experience improved efficiency, better data accuracy, and enhanced accessibility of records (Alshurideh et al., 2022). Digital systems centralize information, making retrieval faster and reducing redundancy in documentation.

Web-based Systems

Web-based systems are widely adopted in educational institutions to enhance data management and service accessibility. Research shows that web-based information systems enable real-time updates, centralized databases, and improved coordination among users (Dwivedi et al., 2021). These systems promote transparency by organizing records into structured databases that can be searched, filtered, and monitored efficiently.

Unified Theory of Acceptance and Use of Technology (UTAUT)

User acceptance plays a significant role in successful system implementation. Contemporary research based on the Unified Theory of Acceptance and Use of Technology (UTAUT) explains that performance expectancy, effort expectancy, and facilitating conditions significantly influence users' intention to adopt new technologies (Tamilmani et al., 2021). This suggests that a web-based lost and found system must be functional, simple, and accessible to encourage students and staff to use it consistently.

Security and data protection

Security and data protection are critical in modern web-based applications. Studies on information system security highlight that authentication mechanisms and role-based access controls are essential in protecting sensitive information and preventing unauthorized access (Bada & Nurse, 2020). Verification processes also enhance accountability by ensuring that system transactions are properly validated before approval.

Database Design

Effective database design further improves information retrieval and system usability. Recent research indicates that structured categorization, indexing, and searchable databases significantly reduce the time required to locate information within large datasets (Alshurideh et al., 2022). These principles support the integration of login authentication, categorized listings,

search functionality, and claim verification mechanisms in web-based systems such as iFind.

Related Studies

Several recent studies have examined the implementation of web-based systems to replace manual institutional processes. Research on digital service platforms in educational settings demonstrates that centralized web-based applications improve operational efficiency, reduce administrative workload, and enhance service delivery (Dwivedi et al., 2021). These systems provide structured monitoring of records and allow users to access services more conveniently compared to manual methods. Studies focusing on campus information systems also show that searchable databases significantly improve data retrieval speed and accuracy. Digital platforms reduce human error and enhance documentation reliability, making them suitable for record management in schools and universities (Alshurideh et al., 2022). These findings support the development of a structured and searchable lost and found system.

In the Philippines, educational institutions have increasingly adopted web-based systems for administrative and academic record management. Recent local research indicates that digital record systems improve accessibility, minimize data loss, and enhance operational monitoring within schools (Garcia

& Lopez, 2021). These systems reduce reliance on paper-based documentation and provide more efficient service processes.

Furthermore, school information systems further reveal that web-based platforms improve response time, data organization, and user satisfaction among students and staff (Santos et al., 2022). Although these studies focus on broader school management systems, their findings highlight the benefits of implementing structured digital platforms within educational institutions.

International research provides additional evidence regarding the effectiveness of web-based institutional systems. Studies conducted in higher education environments demonstrate that online service platforms improve record management efficiency, reduce manual workload, and enhance transparency (Dwivedi et al., 2021). Centralized databases allow institutions to track transactions systematically and ensure accountability.

In addition, research in information system implementation confirms that integrating authentication and structured verification mechanisms enhances system security and user trust (Bada & Nurse, 2020). These findings support the development of a secure and organized web-based lost and found system such as iFind.

Synthesis

The reviewed literature and studies indicate that manual record-keeping systems are inefficient and prone to errors, while web-based information systems improve data organization, accessibility, and operational efficiency. Research emphasizes the importance of usability, user acceptance, security mechanisms, and structured database management in ensuring successful system implementation.

Both local and foreign studies confirm that centralized and searchable digital platforms enhance institutional processes and reduce administrative burden. However, limited studies specifically address a secure, claim-verified lost and found system within a secondary school setting evaluated using recognized software quality standards. Therefore, the development of iFind: A Web-Based Lost and Found System for Colegio de San Jose addresses this gap by providing a structured, secure, and quality-evaluated platform for managing lost and found items.

CHAPTER 3

METHODOLOGY

Research Design

The study used a quantitative developmental research design to develop and evaluate the iFind: Web-Based Lost and Found System for Colegio de San Jose. The developmental approach was used because the study focused on designing, building, and testing a functional web-based system.

A quantitative method was applied in evaluating the system using a structured questionnaire based on selected ISO/IEC 25010 software quality standards. The evaluation focused on the following characteristics: functional suitability, usability, reliability, performance efficiency, and security. These standards were used to measure the overall quality and acceptability of the system.

After the development phase, selected respondents were allowed to use the system and explore its features, such as submitting lost or found reports, browsing item listings, confirming found items, and claiming reported items. The respondents then evaluated the system using a five-point Likert scale. The

collected data were analyzed using descriptive statistics, specifically the mean and standard deviation, to determine the level of acceptability of the system.

Respondents of the Study

The respondents of this study consisted of 20 selected individuals from Colegio de San Jose, 10 IT experts and 10 common users. The respondents were selected using purposive sampling, ensuring that participants were capable of evaluating the system appropriately. Students were selected as respondents because they serve as the primary users of the system, responsible for reporting lost items, confirming found items, and claiming their belongings. Meanwhile, IT experts were included in the study due to their technical knowledge and their role in monitoring reports and verifying item claims through the system's administrative interface. All respondents were willing participants and were available during the testing period. Before participation, they were informed about the purpose of the study and assured that their responses would remain confidential. Their feedback was essential in determining whether the iFind system meets acceptable standards based on ISO/IEC 25010 quality characteristics.

Data Gathering Instrument

The data-gathering instrument used in this study was a system evaluation questionnaire adapted by the researcher based on ISO/IEC 25010 software quality standards. The questionnaire was designed to measure the system in terms of the following criteria; Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The questionnaire was distributed to the respondents after they had used the system. They were instructed to explore the system's features, including logging in, submitting reports for lost or found items, browsing and searching categorized items, uploading images, submitting claim verification proof, claiming items, and evaluating the overall system performance. A pilot test was conducted before the actual data collection to ensure clarity and relevance of the questions. The responses gathered were analyzed using descriptive statistical tools, specifically the computation of the mean and standard deviation, to determine the overall level of system acceptability based on ISO/IEC 25010 standards.

Data Collection

The data collection process for the iFind: Lost and Found System was executed in three distinct phases: preparation, system walkthrough, and formal assessment. Initially, a structured evaluation instrument was developed by integrating the software quality characteristics of the ISO/IEC 25010 model with

the behavioral constructs of the Unified Theory of Acceptance and Use of Technology (UTAUT). This instrument was designed to capture quantitative data through a 5-point Likert scale, ranging from "Strongly Disagree" (1) to "Strongly Agree" (5), allowing for a statistical measure of the evaluators' perceptions.

During the implementation phase, selected evaluators were given controlled access to the iFind platform. To ensure a comprehensive evaluation, participants were required to perform specific functional tasks, such as reporting lost items, posting found items, searching through existing records, and managing item information. This hands-on interaction ensured that the evaluators' subsequent ratings were based on direct experience with the system's performance, usability, and security features.

Following the system exploration, the evaluators completed the structured checklist. The collected data focused on seven key ISO/IEC 25010 criteria: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, and Maintainability. The resulting data were then tabulated and subjected to statistical treatment, specifically the weighted mean, to interpret the system's overall quality and readiness for deployment.

Testing and Evaluation

The study implemented Black Box Testing to evaluate the functionality and usability of the developed system. In this method, the system was tested

based on its inputs and outputs without examining the internal code or program structure. The objective was to determine whether the system performs its intended functions according to the specified requirements.

A total of twenty (20) participants were involved in the testing process, consisting of ten (10) IT experts and ten (10) common users. The IT experts were selected to assess the technical functionality and overall system performance, while the common users represented the actual end-users who interact with the system in real-life scenarios.

During the testing process, the participants were asked to perform specific tasks such as logging into the system, submitting reports for lost or found items, browsing categorized listings, searching for items, uploading images, submitting claim verification proof, and claiming items. After completing these tasks, the participants evaluated whether the system responded correctly to their inputs and whether the features functioned as expected. The results of the testing were then used to determine the reliability and functionality of the developed system.

The system was evaluated based on the evaluation criteria of the standards of ISO 25010 in terms of Functionality Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The researchers will use Purposive sampling which is a form of non-probability sampling technique that makes the researchers rely on their judgment in choosing members of the population to participate in the study.

The results of the survey were interpreted using the scale presented in Table 1. Mean scores were interpreted as follows: 1.00–1.80 as Dissatisfied, 1.81–2.60 as Fair, 2.61–3.40 as Good, 3.41–4.20 as Very Good, and 4.21–5.00 as Very Satisfied. The scale was used to determine the overall evaluation of the system based on the responses of the participants.

Table 1
Likert's Rating Scale for iFind: Lost and Found System

Mean	Description
4.21 - 5.00	Very Satisfied
3.21 - 4.20	Very Good
2.61 - 3.40	Good
1.81 - 2.60	Fair
1.00 - 1.80	Dissatisfied

The obtained results will help the researchers to determine the level of performance and effectiveness of iFind: Lost and Found System in aiming for its objectives. The findings will provide valuable insights for the researchers to identify areas of improvement and make informed decisions.

Data Analysis

The collected responses from the evaluators were tabulated and analyzed using descriptive statistical methods. The Mean was used to determine the average evaluation score of each criterion in the checklist. This helped identify the overall level of system quality and acceptability based on the ISO/IEC 25010

characteristics. A higher mean score indicates a better evaluation of the system's performance and quality.

Meanwhile, the Standard Deviation was used to determine the variability or spread of the evaluators' responses from the average score. This statistical measure helped identify whether the responses of the evaluators were consistent or varied significantly. A lower standard deviation indicates that the evaluators had similar perceptions of the system's quality, while a higher value suggests greater variation in their evaluations.

Software Development Process

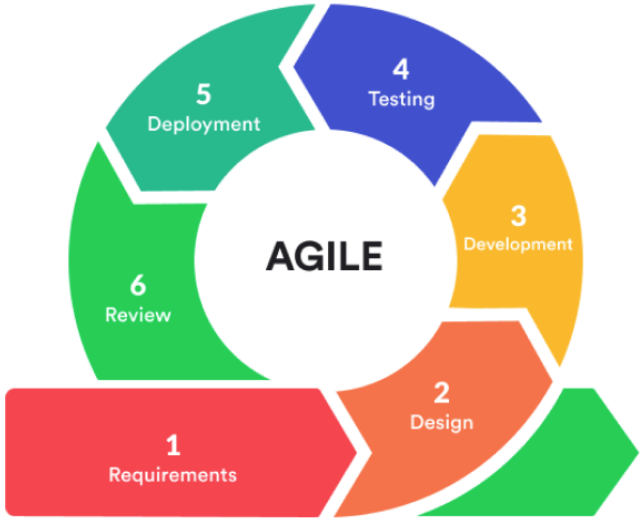


Figure 2. Agile Development Model Diagram

The development of the iFind system followed the Agile Development Model. Agile was used because it allows the system to be developed in small, continuous improvements rather than building everything at once. The system was developed in cycles, where features were created, tested, and improved step by step.

Planning and Requirement Gathering. During this phase, the researchers identified the limitations of the existing manual lost and found process. Students needed to personally visit the school office to inquire about

lost items, which was time-consuming and inefficient. Based on this analysis, the researchers planned to develop a web-based system that allows users to submit lost or found item reports and browse listings online. An administrative page was also planned to manage, approve, and update item records.

Sprint 1 – Basic System Setup. In the first sprint, the basic website structure was developed. This included the homepage, navigation menu, and database structure for storing item details. The public browsing interface was also created to display reported items. At the same time, the admin login page and dashboard structure were developed to allow secure access for administrators. Initial testing was conducted to ensure proper data storage and retrieval.

Sprint 2 – Core Functional Features. In the second sprint, the core system features were implemented. These included: User login system with administrator- controlled account creation, Lost item submission form, found item submission form, Image upload feature, Category selection for organizing items, Search functionality, for faster item retrieval, Browse lost reports page, Browse found reports page, Claim form with proof submission requirement. Each feature was tested after implementation to ensure proper functionality, secure access, and accurate data processing.

Sprint 3 – Status Update and System Improvement. During the third sprint, the status management and claim verification features were finalized. Administrators were given the ability to review submitted claim proofs before

approving item claims. Item statuses could be updated to Pending, Lost, Found, or Claimed through the admin dashboard. Security improvements were applied to ensure that only authenticated users could submit reports and claims. Interface enhancements were also implemented to improve usability, navigation, and overall user experience. Adjustments were applied based on testing feedback to ensure alignment with ISO/IEC 25010 quality standards.

Continuous Testing and Final Implementation. Testing was performed after each sprint to identify and correct errors immediately. Both the public user interface and the admin dashboard were evaluated to ensure smooth functionality. After final refinements, the system was deployed for testing by selected students and staff. The Agile development approach allowed the researchers to continuously improve the system while ensuring alignment with user needs and ISO/IEC 25010 quality standards.

CHAPTER 4

RESULTS AND DISCUSSIONS

This chapter presents the developed iFind: Web-Based Lost and Found System and discusses its features and functionalities. It also presents the results of the system evaluation conducted among selected students and staff of Colegio de San Jose.

The development of the iFind system was guided by the following objectives:

- manage user accounts;
- manage lost and found items;
- create admin notification messages;
- generate reports; and
- evaluate the system using ISO 25010.

iFind: Lost and Found System Homepage

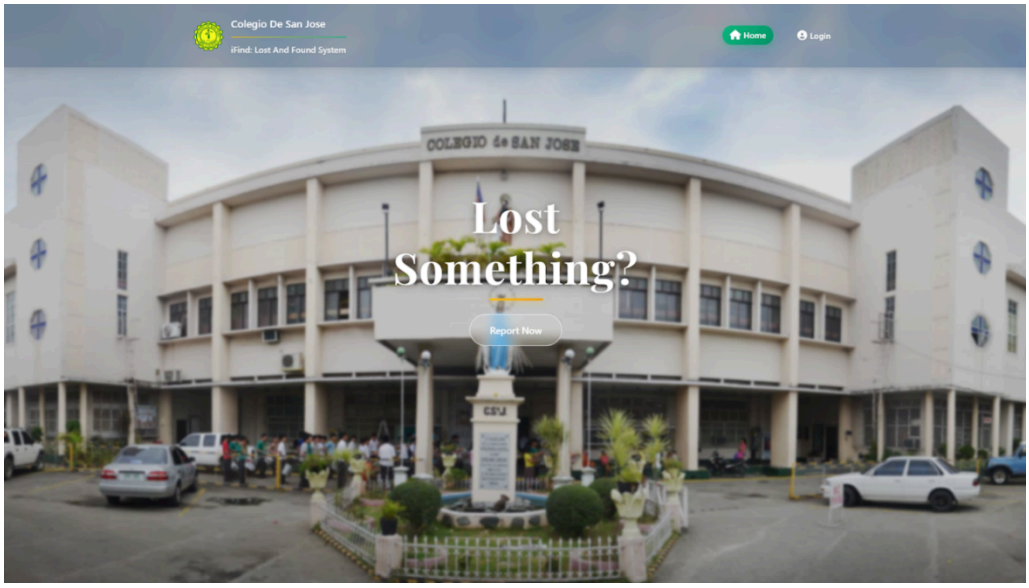


Figure 3. iFind Homepage

Figure 3. shows the homepage of the iFind: Lost and Found System. The homepage serves as the main entry point of the system. It displays the name of the school and the system title to clearly identify the platform. At the top navigation bar, users can access the Home page and Login page. The login feature ensures that only registered users can submit reports and claim items, which enhances system security and accountability. The homepage includes a clear call-to-action button labeled “Report Now,” which directs users to the report submission page. This design allows users to quickly proceed with reporting lost or found items without confusion. The simple layout, readable text, and organized navigation follow usability principles and support ISO/IEC 25010 quality characteristics, particularly usability and functional suitability.

Manage User Accounts

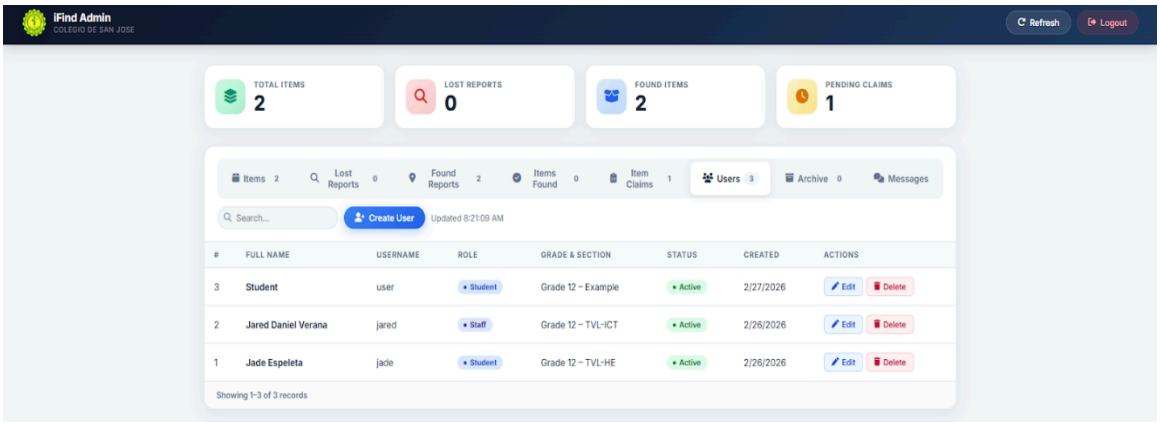


Figure 4. Admin dashboard, User management page

Figure 4 shows the User Management page in the Administrator Dashboard of the iFind system. The interface allows the administrator to manage user accounts within the system. The page displays a list of registered users, including their full name, username, role (Student or Staff), grade and section, account status, and date created. The administrator can create new user accounts using the “Create User” button and can also edit or delete existing accounts. The system includes role-based access control, ensuring that only authorized users can access specific system features. The account status indicator (Active) shows whether a user is currently allowed to log in and use the system. This feature supports the specific objective of managing user accounts and strengthens system security and access control. It also aligns with ISO/IEC 25010 quality characteristics, particularly Security, Functional Suitability, and Maintainability, by ensuring controlled access and proper user management.

Manage lost and found items

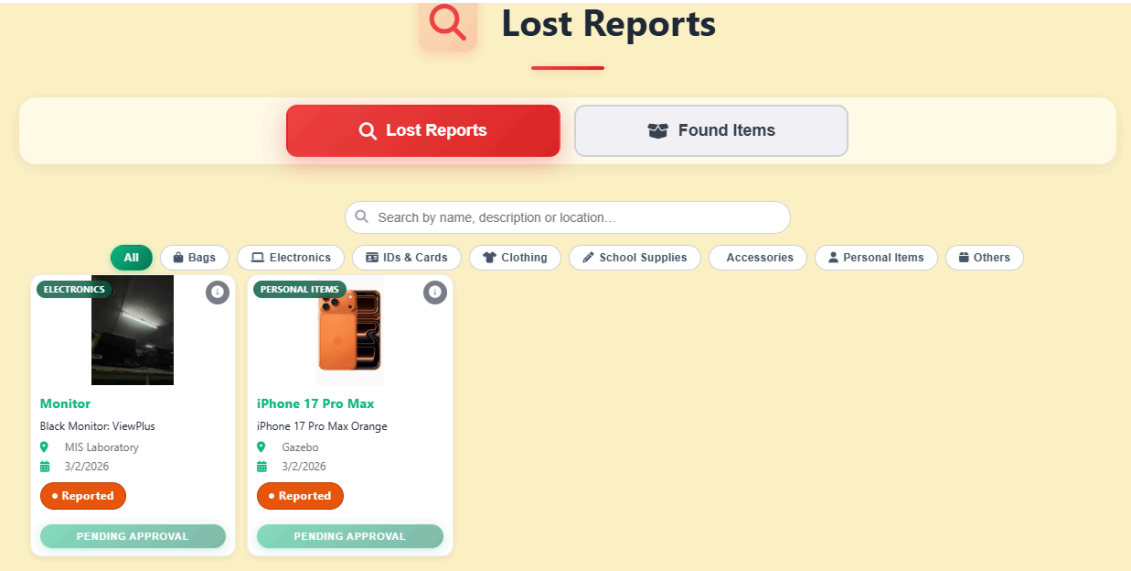


Figure 5. Browse Lost Reports Page

Figure 5 shows the Lost Reports page of the iFind system. This page displays items that have been reported as lost by registered users. Similar to the Found Items page, it includes a search bar and category filters to improve accessibility and user experience. Each item card contains the item image, description, location where it was lost, date reported, and approval status. The clear display of item status (Pending Approval, Reported) ensures transparency and allows users to track their submitted reports. This feature enables proper organization of lost items and supports the system objective of managing lost and found records efficiently.

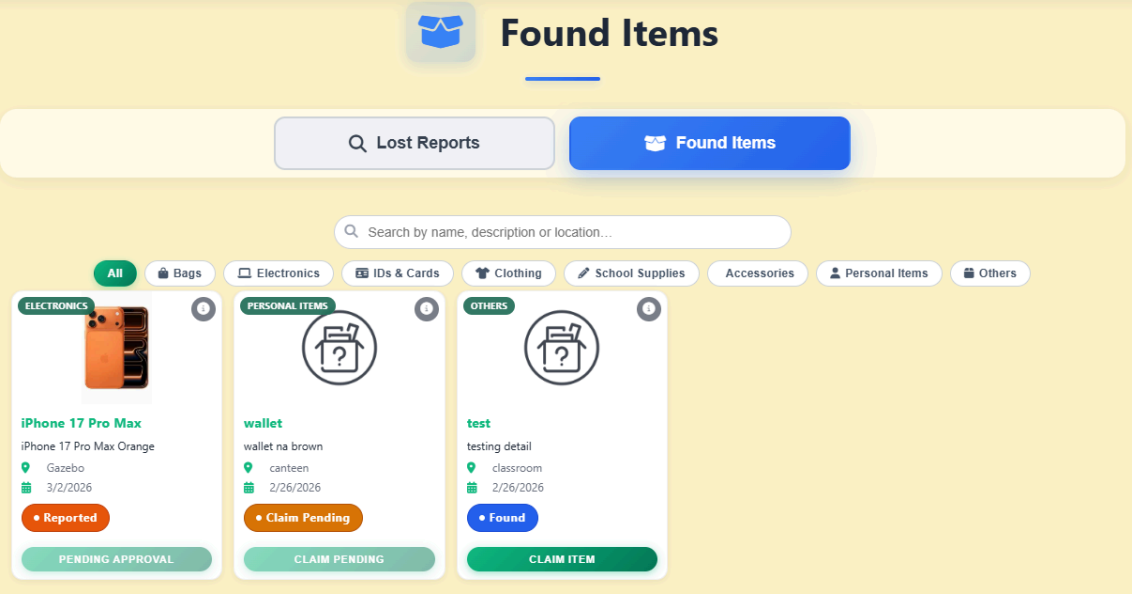


Figure 6. Browse Found Items Page

Figure 6 shows the Found Items page of the iFind system. This interface displays all reported found items that have been submitted and approved by the administrator. The page includes a search bar that allows users to search for items by name, description, or location. It also provides category filters such as Bags, Electronics, IDs & Cards, Clothing, School Supplies, Accessories, Personal Items, and Others. These filters help users quickly narrow down results and locate items efficiently. Each item card displays the item image, name, description, location found, date reported, and current status (Reported, Found, Claim Pending). If the item is available, users may click the “Claim Item” button to submit a claim request. This feature supports functional suitability and usability under ISO/IEC 25010 by ensuring that users can easily browse and locate found items.

Create Admin Notification Messages

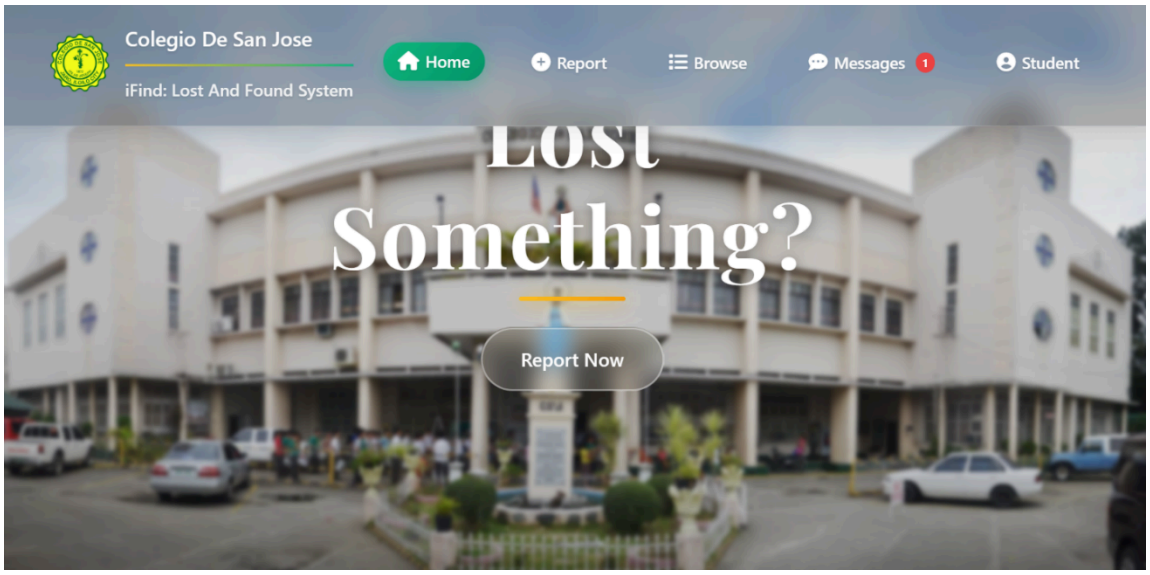


Figure 7. Message Notification

Figure 7 shows the message notification feature of the iFind system. The notification badge displayed beside the Messages icon indicates the number of unread messages. This feature alerts the administrator or user when a new message is received. The notification system ensures that important communications, such as claim concerns or inquiries, are not overlooked. This supports the objective of creating administrator notification messages and enhances responsiveness within the system. This feature aligns with ISO/IEC 25010 quality characteristics, particularly usability and reliability, by ensuring timely awareness of new messages.

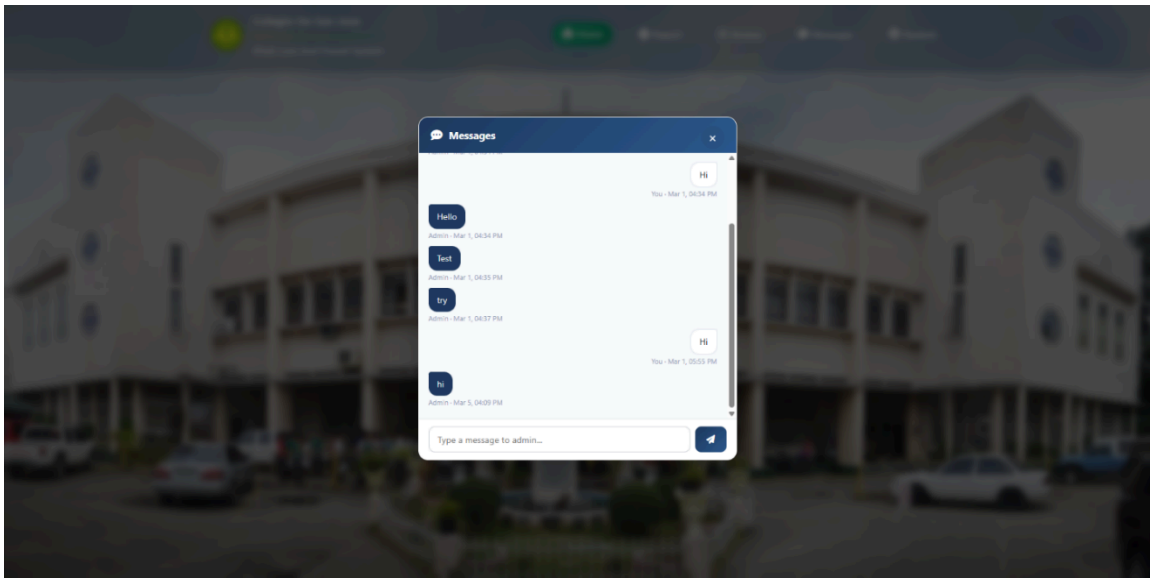


Figure 8. User Messages Inbox

Figure 8 presents the user-side messaging interface. This feature allows registered users to send messages directly to the administrator regarding lost or found items. The interface displays message history, timestamps, and sender identification to maintain proper documentation. Users can type messages and receive replies from the administrator within the same conversation window. This feature improves communication efficiency and transparency between users and the administrator. It supports functional suitability by providing a direct communication channel within the system.

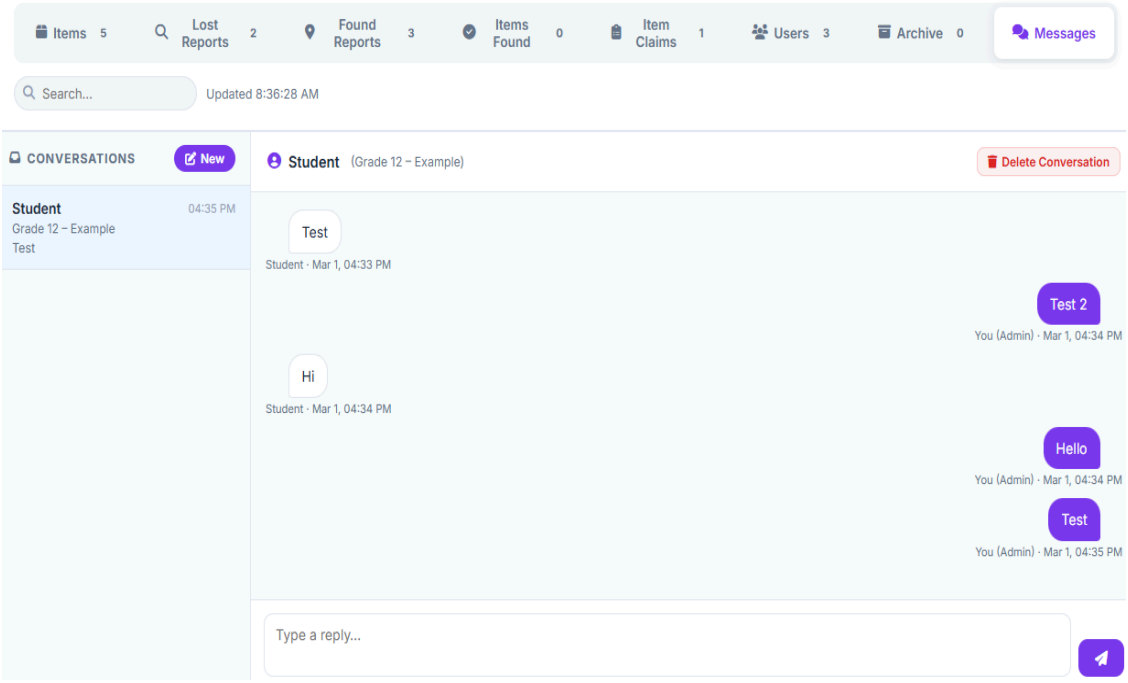


Figure 9. Admin Messages Inbox

Figure 9 shows the administrator's messaging inbox. The administrator can view all user conversations, reply to inquiries, and manage message threads. The interface includes conversation lists, sender details, timestamps, and a reply section. This ensures organized communication management and proper record keeping. This functionality strengthens the system's communication capability and aligns with ISO/IEC 25010 characteristics, particularly usability, reliability, and maintainability.

Generate reports

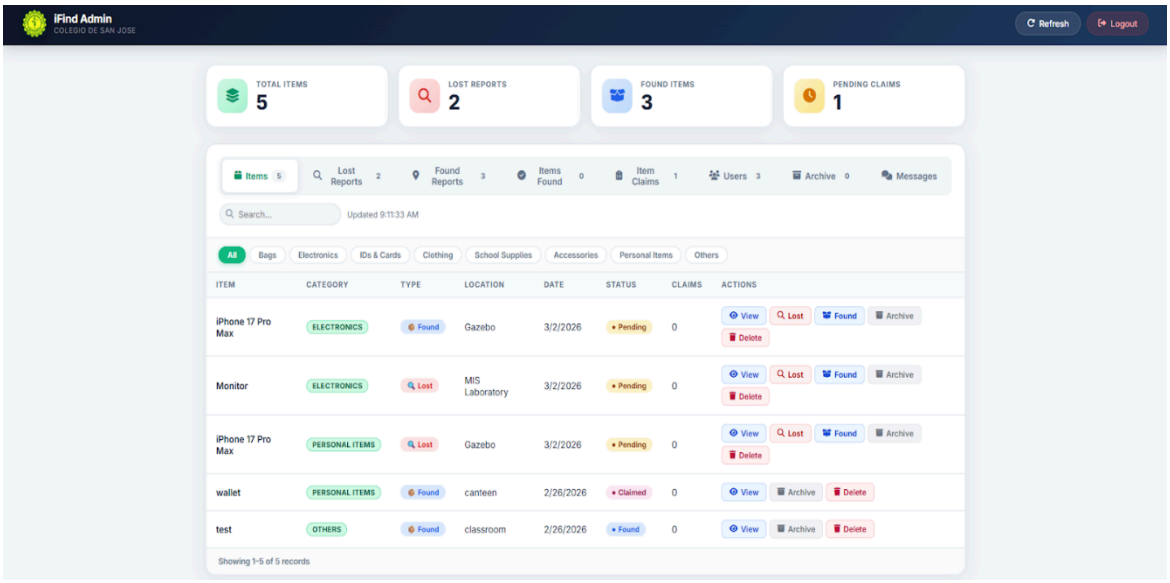


Figure 10. Administrator Dashboard, Item Management and Reports Summary

Figure 10 shows the Administrator Dashboard displaying the item management and report summary interface of the iFind system. This page allows the administrator to monitor and manage all submitted lost and found items. At the top section, summary cards display real-time statistics, including Total Items, Lost Reports, Found Items, and Pending Claims. These summary indicators provide a quick overview of system activity and support efficient monitoring of item records. The table below displays detailed item information, including item name, category, type (Lost or Found), location, date reported, status (Pending, Found, Claimed), and number of claims. The administrator can perform actions such as viewing item details, updating status (Lost or Found), archiving, and deleting records.

System Evaluation Result using ISO 25010

The system was tested using the ISO/IEC 25010 quality characteristics, particularly:

- Functional Suitability
- Performance Efficiency
- Compatibility
- Usability
- Reliability
- Security
- Maintainability
- Portability

After the development and presentation of the iFind system, an evaluation was conducted among 20 selected respondents composed of 10 IT experts and 10 common users. The survey questionnaire used by the researcher was adapted from ISO/IEC 25010 software quality standards. The system was assessed using a five-point Likert scale ranging from 1 (Dissatisfied) to 5 (Very Satisfied). The following sections present the results of the system evaluation in terms of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Table 1
Evaluation Results for Functional Suitability

Criteria in Functional Suitability	Mean	Interpretation	Standard Deviation
Functional Completeness	4.60	Excellent	0.99
Functional Correctness	4.85	Excellent	0.37
Functional Appropriateness	4.65	Excellent	0.75
Average Mean	**4.70**	**Excellent**	

Legend: 4.21 – 5.00 (Excellent) 3.41 – 4.20 (Good) 2.61 – 3.40 (Fair) 1.81 – 2.60 (Average) 1.00 – 1.80 (Poor)

Table 1 displays the tabulated results of the criteria of Functional Suitability, which includes Functional Completeness, Functional Correctness, and Functional Appropriateness. The table presents the evaluation results of the iFind: Web-Based Lost and Found System, focusing on functional suitability, based on feedback from the respondents. The mean score of 4.70 indicates an Excellent level of functional suitability. This suggests that the system effectively provides the required functions and performs them accurately in managing lost and found items within the institution.

Table 2
Evaluation Results for Performance Efficiency

Criteria in Performance Efficiency	Mean	Interpretation	Standard Deviation
Time Behavior	4.8	Excellent	0.41
Resource Utilization	4.6	Excellent	0.6
Capacity	4.7	Excellent	0.57
Total	**4.70**	Excellent	

Legend: 4.21 – 5.00 (Excellent) 3.41 – 4.20 (Good) 2.61 – 3.40 (Fair) 1.81 – 2.60 (Average) 1.00 – 1.80 (Poor)

Table 2 displays the tabulated results of the criteria of Performance Efficiency, which includes Time Behavior, Resource Utilization, and Capacity.

The table presents the evaluation results of the iFind: Web-Based Lost and Found System, focusing on performance efficiency, based on feedback from the respondents. The mean score of 4.70 indicates an Excellent level of performance efficiency. This suggests that the system is capable of performing its functions efficiently while maintaining optimal use of system resources and providing timely responses to user actions.

Table 3.
Evaluation Results for Compatibility

Criteria in Compatibility	Mean	Interpretation	Standard Deviation
Co-existence	4.5	Excellent	1.24
Interoperability	4.7	Excellent	0.47
Total	**4.60**	Excellent	

Legend: *4.21 – 5.00 (Excellent)* *3.41 – 4.20 (Good)* *2.61 – 3.40 (Fair)* *1.81 – 2.60 (Average)* *1.00 – 1.80 (Poor)*

Table 3 displays the tabulated results of the criteria of Compatibility, which includes Co-existence and Interoperability. The table presents the evaluation results of the iFind: Web-Based Lost and Found System, focusing on compatibility, based on feedback from the respondents. The mean score of 4.60 indicates an Excellent level of compatibility. This suggests that the system can operate effectively alongside other applications and systems without causing conflicts while also being capable of interacting with other components when necessary.

Table 4
Evaluation Results for Usability

Criteria in Usability	Mean	Interpretation	Standard Deviation
Appropriateness Recognizability	4.90	Excellent	0.31
Learnability	4.80	Excellent	0.41
Operability	4.65	Excellent	0.59
User Error Protection	4.40	Very Good	1.14
User Interface Aesthetics	4.25	Very Good	1.12
Accessibility	4.70	Excellent	0.47
User Assistance	4.45	Very Good	1.0
Total	**4.59**	Excellent	

Legend: *4.21 – 5.00 (Excellent)* *3.41 – 4.20 (Good)* *2.61 – 3.40 (Fair)* *1.81 – 2.60 (Average)* *1.00 – 1.80 (Poor)*

Table 4 displays the tabulated results of the criteria of Usability, which includes Appropriateness Recognizability, Learnability, Operability, User Error Protection, User Interface Aesthetics, Accessibility, and User Assistance. The table presents the evaluation results of the iFind: Web-Based Lost and Found System, focusing on usability, based on feedback from the respondents. The mean score of 4.59 indicates an Excellent level of usability. This suggests that the system is easy to understand, learn, and operate by users, while also providing an interface that supports efficient interaction and user guidance.

Table 5
Evaluation Results for Reliability

Criteria in Reliability	Mean	Interpretation	Standard Deviation
Maturity	4.4	Very Good	0.94
Availability	4.8	Excellent	0.41
Fault Tolerance	4.2	Very Good	0.99
Recoverability	3.9	Good	1.06
Total	**4.33**	Very Good	

Legend: 4.21 – 5.00 (Excellent) 3.41 – 4.20 (Good) 2.61 – 3.40 (Fair) 1.81 – 2.60 (Average) 1.00 – 1.80 (Poor)

The Table 5 presents the evaluation results of the iFind: Web-Based Lost and Found System, focusing on reliability, based on feedback from the respondents. The mean score of 4.33 indicates a Very Good level of reliability. This suggests that the system is capable of maintaining stable performance and can continue functioning properly even when unexpected conditions occur.

Table 6
Evaluation Results for Security

Criteria in Security	Mean	Interpretation	Standard Deviation
Confidentiality	4	Very Good	1.37
Integrity	3.5	Good	1.47
Non-repudiation	4.3	Very Good	0.89
Accountability	4.2	Very Good	0.88
Authenticity	3.9	Good	1.23
Total	**3.98**	Good	

Legend: 4.21 – 5.00 (Excellent) 3.41 – 4.20 (Good) 2.61 – 3.40 (Fair) 1.81 – 2.60 (Average) 1.00 – 1.80 (Poor)

The Table 6 presents the evaluation results of the iFind: Web-Based Lost and Found System, focusing on security, based on feedback from the respondents. The mean score of 3.98 indicates a Good level of security. This suggests that the system provides acceptable protection of user data and system information, although improvements in certain security features may further enhance the overall security of the system.

Table 7
Evaluation Results for Maintainability

Criteria in Maintainability	Mean	Interpretation	Standard Deviation
Modularity	3.8	Good	1.32
Reusability	4.5	Excellent	0.6
Analyzability	4.4	Very Good	0.99
Modifiability	4.7	Excellent	0.67
Testability	4.7	Excellent	0.66
Total	**4.42**	Very Good	

Legend: *4.21 – 5.00 (Excellent)* *3.41 – 4.20 (Good)* *2.61 – 3.40 (Fair)* *1.81 – 2.60 (Average)* *1.00 – 1.80 (Poor)*

The Table 7 presents the evaluation results of the iFind: Web-Based Lost and Found System, focusing on maintainability, based on feedback from the respondents. The mean score of 4.42 indicates a Very Good level of maintainability. This suggests that the system can be effectively modified, analyzed, and tested when updates or improvements are needed.

Table 8
Evaluation Results for Portability

Criteria in Portability	Mean	Interpretation	Standard Deviation

Adaptability	4.6	Excellent	0.52
Installability	4.5	Excellent	0.67
Replaceability	4.9	Excellent	0.92
Total	**4.67**	Excellent	

Legend: *4.21 – 5.00 (Excellent)* *3.41 – 4.20 (Good)* *2.61 – 3.40 (Fair)* *1.81 – 2.60 (Average)* *1.00 – 1.80 (Poor)*

Table 8 presents the evaluation results of the iFind: Web-Based Lost and Found System, focusing on portability, based on feedback from the respondents. The mean score of 4.67 indicates an Excellent level of portability. This suggests that the system can be easily installed, adapted, and transferred across different environments, making it flexible for implementation and future system upgrades.

Table 9
Overall Evaluation Result for ISO 25010

ISO 25010 Standard	Mean	Interpretation
Functional Suitability	4.7	Excellent
Performance Efficiency	4.7	Excellent
Compatibility	4.6	Excellent
Usability	4.59	Excellent
Reliability	4.33	Very Good
Security	3.98	Good
Maintainability	4.42	Very Good
Portability	4.67	Excellent
Total	**4.50**	Excellent

Legend: *4.21 – 5.00 (Excellent)* *3.41 – 4.20 (Good)* *2.61 – 3.40 (Fair)* *1.81 – 2.60 (Average)* *1.00 – 1.80 (Poor)*

Table 9 presents the overall evaluation of the iFind: Web-Based Lost and Found System based on the ISO/IEC 25010 software quality standards. The

system obtained an overall mean score of 4.50, interpreted as Excellent, indicating that it effectively meets the required software quality attributes. The system received excellent ratings in Functional Suitability (4.70), Performance Efficiency (4.70), Compatibility (4.60), Usability (4.59), and Portability (4.67). Meanwhile, Reliability (4.33) and Maintainability (4.42) were interpreted as Very Good, while Security (3.98) was rated as Good. Overall, the results indicate that the system performs effectively in providing a reliable and efficient platform for managing lost and found items within the institution.

CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

Summary of Findings

The study iFind: Lost and Found System, a platform designed to improve the management of lost and found items within the institution. The system focuses on providing an efficient and organized method of reporting, tracking, and retrieving lost and found items for students, staff, and administrators. The system includes features such as user account management, item reporting, item searching, and administrative management of lost and found records.

Within the system, users are able to report lost items, post found items, and search for items that have been reported within the system. The administrator has the capability to manage reports, verify posted items, and update records to ensure that all information stored in the system remains accurate and organized. The system was designed as a web-based platform, allowing users to access the system through internet-connected devices such as

desktops, laptops, tablets, and mobile phones, making the system convenient and accessible for users within the institution.

The developed system was evaluated by respondents using the ISO/IEC 25010 software quality model, which includes the criteria of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The evaluation was conducted using a structured questionnaire to determine the overall quality and performance of the system.

Based on the evaluation results, the iFind: Web-Based Lost and Found System performed well when assessed in terms of software quality characteristics. The system obtained a mean score of 4.70 in Functional Suitability, 4.70 in Performance Efficiency, 4.60 in Compatibility, and 4.59 in Usability, all interpreted as Excellent. Additionally, the system received a mean score of 4.33 in Reliability and 4.42 in Maintainability, both interpreted as Very Good, while Security obtained a mean score of 3.98, interpreted as Good. Furthermore, Portability obtained a mean score of 4.67, interpreted as Excellent. Overall, the system achieved an overall mean rating of 4.50, interpreted as Excellent, indicating that the developed system meets the required software quality standards and is suitable for managing lost and found items within the institution.

Conclusion

Based on the findings of the study, the following conclusions were drawn:

The development of the iFind Web-Based Lost and Found System was successfully achieved. The system effectively addresses the problems encountered in the manual lost and found process of Colegio de San Jose.

The system provides an organized and centralized platform for reporting and managing lost and found items. It improves efficiency, transparency, and accessibility for both students and administrators.

The Agile SDLC approach contributed significantly to the successful completion of the system by allowing continuous testing and refinement.

The evaluation based on ISO/IEC 25010 standards confirmed that the system meets acceptable software quality requirements. With an overall rating interpreted as Very Satisfactory/Excellent, the system demonstrates strong functionality, usability, reliability, and overall performance.

Therefore, the iFind system is a viable and practical solution that can be implemented in the school to enhance lost and found management.

Recommendations

Based on the evaluation results of the iFind: Lost and Found System, the following recommendations are structured to ensure that the system's high software quality (ISO/IEC 25010) translates into successful long-term adoption (UTAUT).

Students are encouraged to actively adopt the iFind platform as their primary tool for reporting and searching for lost items to build a robust and reliable database. Since the system achieved an "Excellent" rating in Portability and Usability, students should take advantage of its mobile accessibility to report items immediately upon discovery. This proactive engagement will directly enhance the Performance Expectancy of the system, ensuring that the community sees it as a high-value tool for item recovery.

The School Staff are encouraged to act as key influencers by integrating the iFind system into their daily administrative routines. Rather than managing physical lost-and-found collections manually, staff should direct all inquiries to the digital platform. Given the system's high Effort Expectancy (Ease of Use) reflected in its usability scores, staff should provide ongoing feedback to the developers to ensure the interface remains intuitive for users of all technical skill levels, thereby fostering a culture of digital adoption within their respective departments.

The school administration may officially implement the iFind system to modernize the management of lost and found items and reduce reliance on manual record-keeping. By shifting to a centralized, web-based platform, the institution can provide students, staff, and administrators with a more efficient and transparent way to report, track, and retrieve lost belongings. This digital approach ensures that records are accurate, organized, and easily accessible across multiple devices. Administrators will benefit from improved oversight, as

the system allows them to verify posted items, manage reports, and update records in real time. This reduces errors and eliminates the inefficiencies often associated with paper-based processes. With strong ratings in usability, performance, and portability, the system can be smoothly integrated into existing school operations without disrupting daily routines.

The ICT Community and developers should focus on the technical optimization of the platform, specifically addressing the security and reliability metrics. Future iterations should prioritize advanced encryption methods and multi-factor authentication to better protect user data. Additionally, the community should explore the potential for system integration with existing campus databases, such as Student Information Systems, to streamline user verification and enhance the system's overall Compatibility and Functional Suitability.

Future researchers and developers may consider enhancing the iFind system by integrating email or SMS notifications to keep users informed about the status of their reported or found items. This feature would provide real-time updates, ensuring that students and staff are promptly notified when there are changes to their item records, thereby improving communication and user satisfaction. Another potential improvement is the development of a dedicated mobile application version of the system. While the current web-based platform is accessible across devices, a mobile app would offer more convenience and faster access, especially for students and school personnel who rely heavily on

smartphones. Additional features such as QR code verification or ID-based claiming could also be introduced to strengthen security and streamline the process of item retrieval. Future studies on the iFind system may focus on conducting a larger-scale evaluation involving more respondents to further validate its reliability and effectiveness. Expanding the scope of participants would provide a more comprehensive understanding of how the system performs across different user groups, including students, staff, and administrators, over an extended period of use.

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MEMBER: DAUGHTERS OF CHARITY-ST. LOUISE DE MARILLAC EDUCATIONAL SYSTEM

INTEGRATED BASIC EDUCATION DEPARTMENT

February 23, 2025

Noeme M. Jutajero, M.Ed.
Principal
Colegio de San Jose

Dear Ms. Jutajero,

Good day!

We, the Grade 12 TVL-ICT students of Colegio de San Jose, are currently conducting a research study as part of our requirements in Research 2.

In this regard, we respectfully request your permission to conduct our study entitled **"iFind: A Web-Based Lost and Found System for Colegio de San Jose."** The study aims to develop and evaluate a web-based system designed to improve the management of lost and found items within the school.

We would like to request approval to allow selected students and staff to participate in the evaluation of the system by answering a short survey questionnaire based on ISO/IEC 25010 software quality standards. The survey will be conducted within the school premises, and all responses will be treated with confidentiality.

We sincerely hope for your kind approval of this request.

Thank you very much.

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INTEGRATED BASIC EDUCATION DEPARTMENT

Respectfully yours,

Jared Daniel Verana
Ma. Gwen Peñafiel
John Michael Ariar
Kernel Kirck Granpeñas

Noted by:

JOEMAR SALCEDO BRAGAS
Subject Teacher

Approved by:

NOEME M. JUTAJERO
School Principal

QUESTIONNAIRE

iFind: Web Based Lost And Found System

Name (Optional): _____ Signature: _____

Evaluator user-type: IT Expert ( ) User ( )

iFind: Web Based Lost And Found System

Instructions:

Please rate the quality of the system's performance on a scale of 1 to 5, with 5 being Satisfied, 4 Very Good, 3 Good, 2 Fair, and 1 Dissatisfied. Put a check (/) inside the table that corresponds to your rating.

Rating Scale:

Very Satisfied	Very Good	Good	Fair	Dissatisfied
5	4	3	2	1

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A. Functional Suitability

	Indicators	5	4	3	2	1
Completeness	The set of functions covers all the specified tasks and user objectives.					
Correctness	The function provides correct results with the needed degree of precision.					
Appropriatenes s	The function facilitates the accomplishment of specified tasks and objectives.					
	Total					

B. Performance Efficiency

	Indicators	5	4	3	2	1
Time-Behavior	The system responds quickly.					
Resource Utilization	The amounts and types of resources used system when performing its functions meet the requirements					
Capacity	The maximum limits of the products or system parameters meet the requirements.					
Total						

C. Compatibility

	Indicators	5	4	3	2	1
Co-Existence	A system can perform its required functions efficiently while sharing a common environment and resources with other systems, without any harmful impact on any other system					
Interoperability	The system performs its required function efficiently					

Total						

D. Usability

	Indicators	5	4	3	2	1
Appropriatenes s Recognizability	Users can recognize whether a system is appropriate for their needs.					
Learnability	The System Is Easily Managed With Its Labels and Instructions.					
Operability	The system is easy to operate and is kept in a reliable functioning condition.					
User Error	The system protects users against making					
Protection	Errors, such as entering data types that are not suitable for their specific needs.					
User Interface Aesthetics	The user interface enables pleasing and satisfying interactions for the user.					
Accessibility	The system can be used by people with the widest range of characteristics and capabilities.					
Total						

E. Reliability

	Indicators	5	4	3	2	1
Maturity	The system can maintain its performance even when the data volume is increased					
Availability	The System Operates And Is Accessible When required to use.					
Fault Tolerance	The system is capable of handling errors.					
Recoverability	The system can re-establish or recover data in case of failure.					
Total						

F. Security

	Indicators	5	4	3	2	1
Confidentiality	The System Secures Its Information And Property.					
Integrity	The system is designed to prevent unauthorized access to user information.					
Non-repudiation	The system maintains a record process to prevent it from being repudiated.					
Accountability	The system entity is unique.					
Authenticity	The system can verify that the subject's identity is the one claimed by them.					
Total						

G. Maintainability

	Indicators	5	4	3	2	1
Analyzability	The system can detect deficiencies and identify parts that need modification, such as incorrect data entry.					
Modularity	The system component has minimal impact on other components.					
Reusability	The system uses an asset on the other system.					
Modifiability	The System Can Be Effectively And Efficiently Modified.					
Testability	The System Can Be Effectively And Efficiently Modified.					
Total						

H. Portability

	Indicators	5	4	3	2	1

Adaptability	The System Is Operable On Another Operating system					
Installability	The System Can Be Installed From One environment to another					
Replaceability	The System Adapts To Different Environments without accepting other functions					
Total						